EE 66 Fall 2026 — Signals, Dynamics, and Information — Homework 9

This homework is due Friday, October 30, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned). as well as your IPython notebook saved as a PDF.

On occasion, a problem set contains one or more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

1. A Change of Clock

Alok and Baris are watching a cart rolling along a straight track. Its position in meters (y_j) is measured at the times $t_j = j$ seconds ($j = 1,2,3,4,5$):

time t_j (s)	1	2	3	4	5
position y_j (m)	1	3	2	5	4

Alok and Baris measure the time with their own stopwatches, then fit a straight line to their data. Alok records the times above: $\mathbf{t} = (t_1,\ldots,t_5)^\top$ and $\mathbf{y} = (y_1,\ldots,y_5)^\top$. Let $\mathbf{1} = (1,1,1,1,1)^\top$.

Alok then fits the model $y \approx c_0 + c_1 t$ by least squares, minimizing $\|\mathbf{Ac}-\mathbf{y}\|^2$, where $\mathbf{A} = \begin{bmatrix}\mathbf{1} & \mathbf{t}\end{bmatrix}$ and $\mathbf{c} = (c_0,c_1)^\top$.

Baris started his stopwatch 3 seconds late (he reads $s = t-3$). Baris fits the model $y \approx d_0 + d_1 s$ to his measurements by minimizing $\|\mathbf{Bd}-\mathbf{y}\|^2$, where $\mathbf{B} = \begin{bmatrix}\mathbf{1} & \mathbf{s}\end{bmatrix}$, $\mathbf{s} = \mathbf{t} - 3\cdot\mathbf{1}$, and $\mathbf{d} = (d_0,d_1)^\top$.

(a) Determine Baris's least-squares estimates d_0 and d_1. You should not need to invert any non-diagonal matrix.

Solution: Here $\mathbf{s} = (-2,-1,0,1,2)^\top$, so $\mathbf{1}^\top\mathbf{s} = 0$ and the columns of $\mathbf{B}$ are orthogonal:

$$\mathbf{B}^\top\mathbf{B} = \begin{bmatrix}5 & 0\\ 0 & 10\end{bmatrix}, \qquad \mathbf{B}^\top\mathbf{y} = \begin{bmatrix}1+3+2+5+4\\ -2-3+0+5+8\end{bmatrix} = \begin{bmatrix}15\\ 8\end{bmatrix}.$$

The normal equations $\mathbf{B}^\top\mathbf{Bd} = \mathbf{B}^\top\mathbf{y}$ are diagonal, so

$$d_0 = \frac{15}{5} = 3, \qquad d_1 = \frac{8}{10} = 0.8.$$

(b) Explain why Alok and Baris obtain the same fitted vector $\widehat{\mathbf{y}}$. Use this to show that

$$c_1 = d_1 \qquad \text{and} \qquad c_0 = d_0 - 3d_1,$$

and hence determine Alok's estimates c_0 and c_1.

Solution: Since $\mathbf{s} = \mathbf{t} - 3\cdot\mathbf{1}$,

$$\mathbf{B} = \mathbf{A}\begin{bmatrix}1 & -3\\ 0 & 1\end{bmatrix},$$

and this 2×2 matrix is invertible. Hence $\mathrm{Col}(\mathbf{A}) = \mathrm{Col}(\mathbf{B})$. Both students project $\mathbf{y}$ onto the same subspace, and the projection is unique, so they obtain the same $\widehat{\mathbf{y}}$.

Because the columns of $\mathbf{A}$ are linearly independent, $\widehat{\mathbf{y}}$ has a unique representation in terms of $\mathbf{1}$ and $\mathbf{t}$:

$$c_0\mathbf{1} + c_1\mathbf{t} = \widehat{\mathbf{y}} = d_0\mathbf{1} + d_1(\mathbf{t} - 3\cdot\mathbf{1}) = (d_0 - 3d_1)\mathbf{1} + d_1\mathbf{t}.$$

Matching coefficients gives $c_1 = d_1$ and $c_0 = d_0 - 3d_1$. Therefore

$$c_1 = 0.8, \qquad c_0 = 3 - 3(0.8) = 0.6.$$

(c) Let $\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}$ denote the residual.

i. Use the normal equations to explain why $\mathbf{e} \perp \widehat{\mathbf{y}}$.
Solution: The normal equations $\mathbf{A}^\top\mathbf{A}\mathbf{c} = \mathbf{A}^\top\mathbf{y}$ say $\mathbf{A}^\top(\mathbf{y} - \mathbf{A}\mathbf{c}) = \mathbf{A}^\top\mathbf{e} = \mathbf{0}$. Since $\widehat{\mathbf{y}} = \mathbf{A}\mathbf{c}$,

$$\widehat{\mathbf{y}}^\top\mathbf{e} = \mathbf{c}^\top\mathbf{A}^\top\mathbf{e} = \mathbf{c}^\top\mathbf{0} = 0.$$

ii. Determine the minimum squared error $\|\mathbf{e}\|^2$. You should not need to compute the individual residuals.
Solution: Since $\mathbf{y} = \widehat{\mathbf{y}} + \mathbf{e}$ with $\mathbf{e} \perp \widehat{\mathbf{y}}$, the Pythagorean theorem gives

$$\|\mathbf{e}\|^2 = \|\mathbf{y}\|^2 - \|\widehat{\mathbf{y}}\|^2.$$

We have $\|\mathbf{y}\|^2 = 1 + 9 + 4 + 25 + 16 = 55$. Using Baris's orthogonal columns, $\widehat{\mathbf{y}} = d_0\mathbf{1} + d_1\mathbf{s}$ and

$$\|\widehat{\mathbf{y}}\|^2 = d_0^2\|\mathbf{1}\|^2 + d_1^2\|\mathbf{s}\|^2 = 9\cdot 5 + 0.64\cdot 10 = 51.4.$$

Therefore $\|\mathbf{e}\|^2 = 55 - 51.4 = 3.6$.

(d) Now suppose the measurement times $t_1, \ldots, t_m$ are arbitrary, but not all equal, with mean $\bar{t} = \frac{1}{m}\sum_{j=1}^m t_j$; let $\bar{y} = \frac{1}{m}\sum_{j=1}^m y_j$. Show that the least-squares line $y \approx c_0 + c_1 t$ passes through the point $(\bar{t}, \bar{y})$.
Solution: The first row of the normal equations $\mathbf{A}^\top(\mathbf{y} - \mathbf{A}\mathbf{c}) = \mathbf{0}$ is

$$\mathbf{1}^\top(\mathbf{y} - c_0\mathbf{1} - c_1\mathbf{t}) = 0 \quad \Longrightarrow \quad \sum_{j=1}^m y_j = mc_0 + c_1\sum_{j=1}^m t_j.$$

Dividing by m gives $\bar{y} = c_0 + c_1\bar{t}$, so the point $(\bar{t}, \bar{y})$ lies on the least-squares line. (The times are not all equal, so $\mathbf{A}$ has linearly independent columns and the least-squares solution is unique.)

2. SVD Computation

(a) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 5 \\ 3 & 1 & -1 \\ 2 & -1 & 4 \end{bmatrix}.$$

Observe that the columns of matrix $\mathbf{A}$ are mutually orthogonal with norms $\sqrt{14}, \sqrt{3}, \sqrt{42}$.

Verify numerically that columns $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ **and** $\begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix}$ **are orthogonal to each other.**

Solution: Taking the inner product of the two vectors, we have

$$\left\langle \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix} \right\rangle = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}^\top \begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix} = 5 - 1 - 4 = 0. \tag{1}$$

So the two columns are orthogonal to each other.

(b) **Write $\mathbf{A} = \mathbf{BD}$, where $\mathbf{B}$ is an orthonormal matrix and $\mathbf{D}$ is a diagonal matrix.** What is $\mathbf{B}$? What is $\mathbf{D}$?

Solution: We compute the norm for each column and divide each column by its norm to obtain matrix $\mathbf{B}$. Matrix $\mathbf{D}$ is formed by placing the norms on the diagonal.

$$\mathbf{B} = \begin{bmatrix} -\frac{1}{\sqrt{14}} & \frac{1}{\sqrt{3}} & \frac{5}{\sqrt{42}} \\ \frac{3}{\sqrt{14}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{42}} \\ \frac{2}{\sqrt{14}} & -\frac{1}{\sqrt{3}} & \frac{4}{\sqrt{42}} \end{bmatrix} \tag{2}$$

$$\mathbf{D} = \begin{bmatrix} \sqrt{14} & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{42} \end{bmatrix} \tag{3}$$

(c) **Write out a valid singular value decomposition of $\mathbf{A} = \mathbf{U\Sigma V}^\top$ using the result from the previous part.** Note that the singular values in $\mathbf{\Sigma}$ should be ordered from largest to smallest.
(HINT: There is no need to compute any eigenvalues.)

Solution: Using part b, we can write

$$\mathbf{A} = \mathbf{BD} = \mathbf{BDI} = \begin{bmatrix} -\frac{1}{\sqrt{14}} & \frac{1}{\sqrt{3}} & \frac{5}{\sqrt{42}} \\ \frac{3}{\sqrt{14}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{42}} \\ \frac{2}{\sqrt{14}} & -\frac{1}{\sqrt{3}} & \frac{4}{\sqrt{42}} \end{bmatrix} \begin{bmatrix} \sqrt{14} & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

Reordering the columns and rows of $\mathbf{B}$ and $\mathbf{I}$ so that the diagonal entries of $\mathbf{D}$ are in decreasing order, we have

$$\begin{bmatrix} \frac{5}{\sqrt{42}} & -\frac{1}{\sqrt{14}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{42}} & \frac{3}{\sqrt{14}} & \frac{1}{\sqrt{3}} \\ \frac{4}{\sqrt{42}} & \frac{2}{\sqrt{14}} & -\frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \sqrt{42} & 0 & 0 \\ 0 & \sqrt{14} & 0 \\ 0 & 0 & \sqrt{3} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \tag{5}$$

We now have a SVD of $\mathbf{A}$.

(d) Given a new matrix

$$\mathbf{A} = \frac{1}{\sqrt{50}} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} + \frac{3}{\sqrt{50}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}, \tag{6}$$

write out a singular value decomposition of matrix A in the form $\mathbf{U\Sigma V}^\top$. Remember that the singular values in $\mathbf{\Sigma}$ should be ordered from the largest to smallest.
(HINT: You don't have to compute any eigenvalues for this. Some useful observations are that

$$\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = 0, \quad \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0, \quad \left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = \left\| \begin{bmatrix} -4 \\ 3 \end{bmatrix} \right\| = 5, \quad \left\| \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{2}.$$

)

Solution:

The singular value decomposition can be written in the form

$$\mathbf{A} = \sum_{i=1}^{2} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \tag{7}$$

with unit orthonormal vectors $\{\mathbf{u}_i\}$ and $\{\mathbf{v}_i\}$. From the given observations, we can see that the vectors we were provided are orthogonal, so we can just normalize them to get the desired answer. Taking it step by step:

$$\mathbf{A} = \frac{1}{\sqrt{50}} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} + \frac{3}{\sqrt{50}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \tag{8}$$

$$= \frac{5}{\sqrt{50}} \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} + \frac{3 \cdot 5}{\sqrt{50}} \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \tag{9}$$

$$= \frac{5\sqrt{2}}{\sqrt{50}} \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} + \frac{3 \cdot 5\sqrt{2}}{\sqrt{50}} \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \tag{10}$$

$$= 1 \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} + 3 \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}. \tag{11}$$

From this, we can derive

$$\mathbf{u_1} = \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix}, \mathbf{u_2} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}, \mathbf{v_1} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \mathbf{v_2} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \tag{12}$$

and corresponding singular values $\sigma_1 = 3, \sigma_2 = 1$ because we need to order them by size in decreasing order. This gives the singular value decomposition

$$\mathbf{A} = \begin{bmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}^\top \tag{13}$$

(e) Let us define a new matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 4 \\ 1 & 4 \end{bmatrix}.$$

Find the SVD of A by following the standard algorithm introduced in lecture (i.e. by computing the eigendecomposition of $\mathbf{A}^\top\mathbf{A}$). Also find the eigenvectors and eigenvalues of A. You may use a calculator to solve the roots of the characteristic polynomial *for the eigenvalues of* $\mathbf{A}$, but not for the SVD. Is there a relationship between the eigenvalues or eigenvectors of $\mathbf{A}$ with the SVD of $\mathbf{A}$?

Solution: Since we have a square matrix, we will arbitrarily use $\mathbf{A}^\top\mathbf{A}$ for our SVD:

$$\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 4 & 4 \end{bmatrix}\begin{bmatrix} -1 & 4 \\ 1 & 4 \end{bmatrix} \tag{14}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 32 \end{bmatrix} \tag{15}$$

Next, we find the eigenvalues of the above matrix.

$$\det(\lambda\mathbf{I} - \mathbf{A}^\top\mathbf{A}) = (\lambda - 2)(\lambda - 32) = 0 \tag{16}$$

Hence, the eigenvalues are $\lambda_1 = 32$ and $\lambda_2 = 2$, and the singular values are $\sigma_1 = \sqrt{32} = 4\sqrt{2}$ and $\sigma_2 = \sqrt{2}$. Next, we find the right singular vectors (i.e. the columns of $\mathbf{V}$). Finding $\text{null}(\lambda_1\mathbf{I} - \mathbf{A}^\top\mathbf{A})$ and $\text{null}(\lambda_2\mathbf{I} - \mathbf{A}^\top\mathbf{A})$ will give us $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively.

Hence, $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (the eigenvectors are already normalized here).

Lastly, we find the left singular vectors (the columns of $\mathbf{U}$)

$$\mathbf{u}_1 = \frac{1}{\sigma_1}\mathbf{A}\mathbf{v}_1 \tag{17}$$

$$= \frac{1}{4\sqrt{2}}\begin{bmatrix} -1 & 4 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{18}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \tag{19}$$

Similarly, we get $\mathbf{u}_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.

So the full SVD representation of $\mathbf{A}$ is given below

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}\begin{bmatrix} 4\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{20}$$

Now that we have found the SVD of $\mathbf{A}$, we will find the eigenvalues and eigenvectors of $\mathbf{A}$. Let us start with the eigenvalues:

$$\det(\lambda\mathbf{I} - \mathbf{A}) = (\lambda + 1)(\lambda - 4) - 4 = \lambda^2 - 3\lambda - 8 = 0 \tag{21}$$

Using the quadratic formula, the eigenvalues are $\lambda_1 = \frac{3+\sqrt{41}}{2} \approx 4.7$ and $\lambda_2 = \frac{3-\sqrt{41}}{2} \approx -1.7$.

Since we already used $\mathbf{v}_1, \mathbf{v}_2$ for the SVD, let us denote the eigenvectors of $\mathbf{A}$ as $\mathbf{r}_1, \mathbf{r}_2$.

Finding $\text{null}(\lambda_1\mathbf{I} - \mathbf{A})$ and $\text{null}(\lambda_2\mathbf{I} - \mathbf{A})$ will give us $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively.

Hence, the normalized eigenvectors of $\mathbf{A}$ are $\mathbf{r}_1 \approx \begin{bmatrix} 0.57 \\ 0.82 \end{bmatrix}$ and $\mathbf{r}_2 \approx \begin{bmatrix} -0.98 \\ 0.17 \end{bmatrix}$.

We notice that there is no relationship between the eigenvalues or eigenvectors of $\mathbf{A}$ with the SVD of $\mathbf{A}$.

3. Rank 1 Decomposition and Image Compression With SVD

In this problem, we will introduce an important application of the singular value decomposition (SVD) called low-rank approximation in the context of image compression. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ of rank $r \leq \min\{m,n\}$, we know that we can write $\mathbf{A}$ using the outer-product form of the SVD:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \tag{22}$$

If our matrix is high-rank, i.e., $r \approx \min\{m,n\}$, then almost all the σ_i will be nonzero and non-negligible. However, if the data has some linear, low-rank essential structure, as is usually the case with real data such as images, most of our singular values will be very small (but usually nonzero due to noise or disturbances). If, say, the data has rank ℓ, then the first ℓ singular values are large, and the remaining $r-\ell$ are small:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{23}$$

$$= \sum_{i=1}^{\ell} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top + \sum_{i=\ell+1}^{r} \underbrace{\sigma_i}_{\approx 0} \mathbf{u}_i \mathbf{v}_i^\top \tag{24}$$

$$\approx \sum_{i=1}^{\ell} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \tag{25}$$

This motivates approximating the data as

$$\mathbf{A}_\ell := \sum_{i=1}^{\ell} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \tag{26}$$

and using this compressed data for further analysis. As images are readily represented as matrices, in this problem, we will see how to make use of this low rank approximation technique to compress images without severely deteriorating the image quality.

Throughout this problem we will work in the Jupyter Notebook accompanying this homework. Part 1 of the notebook relates to finding rank 1 decompositions of images, while Parts 2 and 3 focus on image compression. Make sure to read through all the sections of the notebook to understand how we will use what we know about the SVD in order to compress images! You should write all your answers in the written submission and you don't need to submit the parts of the notebook related to this problem.

We will start by decomposing a few images into linear combinations of rank 1 matrices. Remember that outer product of two vectors $\mathbf{s}\mathbf{g}^\top$ gives a rank 1 matrix. It has rank 1 because the column span is one-dimensional — multiples of $\mathbf{s}$ only — and the row span is also one dimensional — multiples of $\mathbf{g}^\top$ only. More generally, this exercise should provide intuition into how the outer product form of the SVD decomposes the underlying matrix into a sum of constituent rank 1 matrices with different weightings given by the singular values.

(a) Consider a standard 4×4 checkerboard shown in Figure 1. Assume that black colors represent -1 and that white colors represent 1.

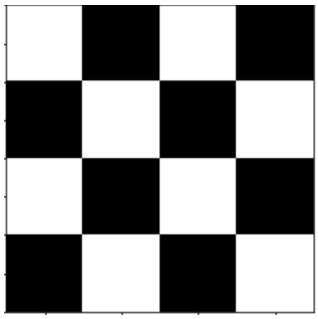


Figure 1: 4×4 checkerboard.

The checkerboard is given by the following 4×4 matrix $\mathbf{L}$:

$$\mathbf{L} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix} \tag{27}$$

Express L as a linear combination of outer products. There are many solutions, but providing one is enough. It is not necessary for the vectors to be normalized. You may use the helper functions in the accompanying Jupyter Notebook to check your answer.

(HINT: In order to determine how many rank 1 matrices you need to combine to represent the matrix, find the rank of the matrix you are trying to represent.)

Solution: The matrix $\mathbf{L}$ only has rank 1, since column vectors 1, 3 are the same, and column vectors 2, 4 are multiples of the first or third column. This means that we can express $\mathbf{L}$ by multiplying the first column vector $\begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix}^\top$ by the multiples required to generate the other columns, which are $1, -1, 1, -1$. As a result, we get the following outer product form:

$$\mathbf{L} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \tag{28}$$

(b) We next consider the same checkerboard given by the following 4×4 matrix $\mathbf{C}$ where black colors now represent 0 and that white colors represent 1.

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \tag{29}$$

Express C as a linear combination of outer products. There are many solutions, but providing one is enough. It is not necessary for the vectors to be normalized. You may use the helper functions in the accompanying Jupyter Notebook to check your answer.

(HINT: Note that $\mathbf{C}$ is a rank 2 matrix and therefore you need 2 outer products to represent it. To find these individual outer products consider the columns of $\mathbf{C}$.)

Solution: We first note that $\mathbf{C}$ is a rank 2 matrix and thus will be a summation of 2 outer products. There are infinitely many outer products that we can choose, but we will just consider one linear

combination. To find this, we note that $\mathbf{C}$ has two distinct columns. The first and third columns are given by $\begin{bmatrix}1\\0\\1\\0\end{bmatrix}$, while the second and fourth columns are $\begin{bmatrix}0\\1\\0\\1\end{bmatrix}$. Therefore

$$\mathbf{C} = \begin{bmatrix}1\\0\\1\\0\end{bmatrix}\begin{bmatrix}1 & 0 & 1 & 0\end{bmatrix} + \begin{bmatrix}0\\1\\0\\1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 1\end{bmatrix}. \tag{30}$$

(c) Now consider the Swiss flag shown in Figure 2. Assume that red colors represent 0 and that white colors represent 1.

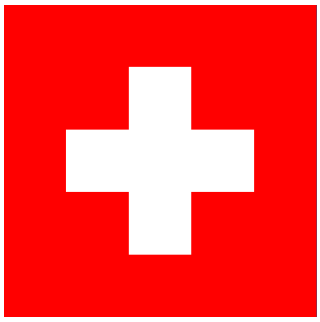


Figure 2: Swiss flag.

Assume that the Swiss flag is given by the following 5×5 matrix $\mathbf{S}$:

$$S = \begin{bmatrix}0 & 0 & 0 & 0 & 0\\0 & 0 & 1 & 0 & 0\\0 & 1 & 1 & 1 & 0\\0 & 0 & 1 & 0 & 0\\0 & 0 & 0 & 0 & 0\end{bmatrix} \tag{31}$$

Express S as a linear combination of outer products. There are many solutions, but providing one is enough. It is not necessary for the vectors to be normalized. You may use the helper functions in the accompanying Jupyter Notebook to check your answer.

Solution: We again note that $\mathbf{S}$ is a rank 2 matrix and will be a summation of 2 outer products. There are again infinite solutions, but we will provide a couple. To see the first of these solutions, we note that we can reconstruct all of the columns of $\mathbf{S}$ using linear combinations of $\begin{bmatrix}0\\0\\1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\\0\\1\\0\end{bmatrix}$. For example, to obtain the third column of $\mathbf{S}$ we can simply scale the aforementioned vectors by 1 and then

sum their contributions. Therefore, we can write $\mathbf{S}$ as an outer product expansion using these vectors:

$$\mathbf{S} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \end{bmatrix}. \tag{32}$$

Similarly, the columns of $\mathbf{S}$ can be written as linear combinations of $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. This gives rise to the outer product summation

$$\mathbf{S} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \end{bmatrix} \tag{33}$$

(d) To represent the full $\mathbf{A} \in \mathbb{R}^{m \times n}$, we need $m \cdot n$ numbers. If we instead take the rank-k approximation $\mathbf{A}_k := \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$, **in terms of** $k, m,$ **and** n**, how many real numbers do we require to describe** $\mathbf{A}_k$**?**
Solution: We require storage of $k(m+n+1)$ real numbers (k singular values, k left singular vectors of size m, and k right singular vectors of size n).

(e) Here we show two images of a kitten: An uncompressed $\mathbf{A}$ with dimensions 900×1200 (right), and a compressed rank 200 approximation of $\mathbf{A}$ (left). **Can you easily spot any difference between the two images even though we are using only about 40% of the data?**

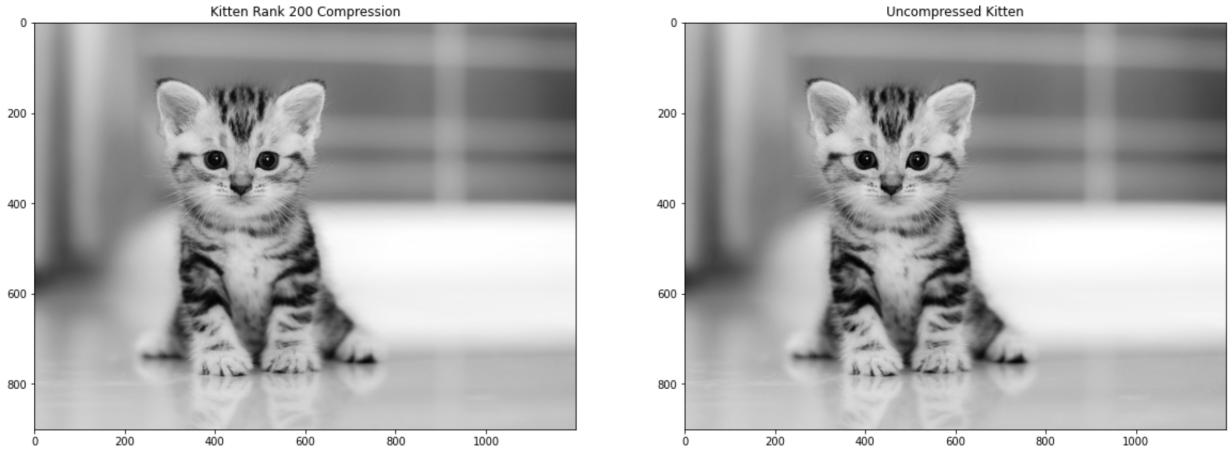


Figure 3: Compressed vs uncompressed image of a kitten.

Solution: We expect that you notice very little to no change between the images. This means we can represent most of the image with only the top 200 singular values!

(f) The accompanying Jupyter Notebook contains a slider that allows you to adjust the rank k of the approximation. **What is the lowest acceptable value of k that you can go without losing too much image quality? What percentage of the data is needed?**

Solution: Playing around with the slider, around $k = 70$ was the lowest we could go without noticing too much image loss. Only about 15% of the data is needed. You just have to be in the ballpark for credit.

4. SVD Proofs

Given the SVD of a matrix $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, we can write $\mathbf{V} = \begin{bmatrix}\mathbf{V}_r & \mathbf{V}_{n-r}\end{bmatrix}$, where $\mathbf{V}_r$ are the first r columns of $\mathbf{V}$ and $\mathbf{V}_{n-r}$ are the last $n-r$ columns of $\mathbf{V}$. Similarly, we have $\mathbf{U} = \begin{bmatrix}\mathbf{U}_r & \mathbf{U}_{m-r}\end{bmatrix}$, where $\mathbf{U}_r$ are the first r columns of $\mathbf{U}$ and $\mathbf{U}_{m-r}$ are the last $m-r$ columns of $\mathbf{U}$.

(a) Show, for a general matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $\text{rank}(\mathbf{A}) = r$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, that $\text{Null}(\mathbf{A}) = \text{Col}(\mathbf{V}_{n-r})$. *HINT: How do we show that two spaces are equal? Consider using the outer product form of SVD. It may also be helpful to recall the rank-nullity theorem:* $\dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A})) = n$.

Solution: $\text{Null}(\mathbf{A}) \subseteq \text{Col}(\mathbf{V}_{n-r})$

We can start by writing the SVD of $\mathbf{A}$ in outer product form:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

Let $\mathbf{x}$ be a (nonzero) vector such that $\mathbf{A}\mathbf{x} = \mathbf{0}$. Then

$$\mathbf{A}\mathbf{x} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \mathbf{x}$$

Here, we require $\mathbf{v}_i^\top \mathbf{x} = 0$ for $i = 1$ to r. Hence, $\mathbf{x}$ must be orthogonal to $\mathbf{v}_i$ for $i = 1$ to r, since each $\sigma_i \neq 0$ and $\mathbf{u}_i$ are linearly independent. Since the columns of $\mathbf{V}$ form a basis for $\mathbb{R}^n$, we can write $\mathbf{x}$ as a linear combination of the columns of $\mathbf{V}$. However, $\mathbf{x}$ is orthogonal to the first r columns, which means that it must be written as a linear combination of the rest of the columns. This means that $\mathbf{x} \in \text{Col}(\mathbf{V}_{n-r})$. Since $\mathbf{x}$ is an arbitrary nullspace vector of $\mathbf{A}$ by construction, it must be the case that $\text{Null}(\mathbf{A}) \subseteq \text{Col}(\mathbf{V}_{n-r})$.

By the rank-nullity theorem, $\dim(\text{Null}(\mathbf{A})) = n - r$ since $\text{rank}(\mathbf{A}) = r$. At this point, we observe that $\dim(\text{Null}(\mathbf{A})) = \dim(\text{Col}(\mathbf{V}_{n-r}))$, and $\text{Null}(\mathbf{A}) \subseteq \text{Col}(\mathbf{V}_{n-r})$. If two spaces share the same dimension, and one space is contained inside of the other, then the two spaces must be the same. Thus, $\text{Null}(\mathbf{A}) = \text{Col}(\mathbf{V}_{n-r})$. Alternatively, you can prove the reverse direction as well:

$\text{Null}(\mathbf{A}) \supseteq \text{Col}(\mathbf{V}_{n-r})$ (Alternative Proof)

If we have an arbitrary $\widetilde{\mathbf{v}} \in \text{Col}(\mathbf{V}_{n-r})$, it must be orthogonal to each of $\mathbf{v}_i$ for $i = 1$ to r. Hence,

$$\mathbf{A}\widetilde{\mathbf{v}} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \underbrace{\mathbf{v}_i^\top \widetilde{\mathbf{v}}}_{=0} = \mathbf{0}$$

Thus, $\widetilde{\mathbf{v}} \in \text{Null}(\mathbf{A})$ and $\text{Null}(\mathbf{A}) \supseteq \text{Col}(\mathbf{V}_{n-r})$.

This completes the proof that $\text{Null}(\mathbf{A}) = \text{Col}(\mathbf{V}_{n-r})$.

(b) Show, for a general matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $\text{rank}(\mathbf{A}) = r$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, that $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U}_r)$.

Solution: $\text{Col}(\mathbf{A}) \subseteq \text{Col}(\mathbf{U}_r)$

We can again start by writing the SVD of $\mathbf{A}$ in outer product form:

$$\mathbf{A} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

We know that $\text{Col}(\mathbf{A}) \coloneqq \{\mathbf{b} : \mathbf{A}\mathbf{x} = \mathbf{b}\}$. Hence, if we multiply $\mathbf{A}$ by $\mathbf{x}$:

$$\begin{aligned}\mathbf{A}\mathbf{x} &= \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top \mathbf{x} \\ &= \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \left(\mathbf{v}_i^\top \mathbf{x}\right) \\ &= \sum_{i=1}^{r} \left(\mathbf{v}_i^\top \mathbf{x}\right) \sigma_i \mathbf{u}_i\end{aligned}$$

which is a linear combination of $\mathbf{u}_1, \ldots, \mathbf{u}_r$. Hence, $\mathbf{A}\mathbf{x} \in \text{Col}(\mathbf{U}_r)$ and so $\text{Col}(\mathbf{A}) \subseteq \text{Col}(\mathbf{U}_r)$.

Since $\dim(\text{Col}(\mathbf{A})) = \dim(\text{Col}(\mathbf{U}_r)) = r$, it must be the case that $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U}_r)$. Alternatively, you can show the other direction as well:

$\text{Col}(\mathbf{A}) \supseteq \text{Col}(\mathbf{U}_r)$ (Alternative Proof)
Suppose $\mathbf{b} \in \text{Col}(\mathbf{U}_r)$, so $\mathbf{U}_r\mathbf{x} = \mathbf{b}$ for some vector $\mathbf{x}$. We can show that $\mathbf{b} \in \text{Col}(\mathbf{A})$. First, consider the compact SVD form of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top$$

where $\mathbf{\Sigma}_r \in \mathbb{R}^{r \times r}$ is diagonal. Now, define the vector $\widetilde{\mathbf{x}} = \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{x}$. Notice that

$$\begin{aligned}\mathbf{A}\widetilde{\mathbf{x}} &= \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top \widetilde{\mathbf{x}} \\ &= \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{x} \\ &= \mathbf{U}_r \mathbf{x} = \mathbf{b}\end{aligned}$$

so $\mathbf{b} \in \text{Col}(\mathbf{A})$, and thus, $\text{Col}(\mathbf{A}) \supseteq \text{Col}(\mathbf{U}_r)$.

This completes the proof that $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U}_r)$.

(c) Show, for a general matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $\text{rank}(\mathbf{A}) = r$ and $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, that $\text{Null}(\mathbf{A}^\top) = \text{Col}(\mathbf{U}_{m-r})$ and $\text{Col}(\mathbf{A}^\top) = \text{Col}(\mathbf{V}_r)$. Use these facts and the previous subparts to show that $\dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A}^\top)) = m$, and that $\text{Col}(\mathbf{A})$ and $\text{Null}(\mathbf{A}^\top)$ are orthogonal.

Solution:
Let us define the SVD of $\mathbf{A}^\top$ as $\mathbf{A} = \widetilde{\mathbf{U}}\widetilde{\mathbf{\Sigma}}^\top\widetilde{\mathbf{V}}^\top$. Observe that this is equal to

$$\mathbf{A}^\top = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top = \mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top$$

By pattern-matching, this implies that $\widetilde{\mathbf{U}} = \mathbf{V}$, $\widetilde{\mathbf{V}} = \mathbf{U}$.
With this in mind, we can immediately argue that $\text{Null}(\mathbf{A}^\top) = \text{Col}(\widetilde{\mathbf{V}}_{m-r}) = \text{Col}(\mathbf{U}_{m-r})$ and $\text{Col}(\mathbf{A}^\top) = \text{Col}(\widetilde{\mathbf{U}}_r) = \text{Col}(\mathbf{V}_r)$.

We know that $m = \dim(\text{Col}(\mathbf{U}_r)) + \dim(\text{Col}(\mathbf{U}_{m-r})) = \dim(\text{Col}(\mathbf{A})) + \dim(\text{Null}(\mathbf{A}^\top))$. Since $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U}_r)$ and $\text{Null}(\mathbf{A}^\top) = \text{Col}(\mathbf{U}_{m-r})$, $\text{Col}(\mathbf{A})$ and $\text{Null}(\mathbf{A}^\top)$ must be orthogonal since all the columns of $\mathbf{U}_r$ and all the columns of $\mathbf{U}_{m-r}$ are orthogonal.

5. Conceptual PCA

(a) Consider a data matrix $\mathbf{A} \in \mathbb{R}^{d \times n}$, where n is the number of data points and d is the dimensionality of each data point. The data points, $\mathbf{a}_i$, are the columns of $\mathbf{A}$. One way to formulate PCA is to find an ℓ-dimensional subspace that minimizes the total squared reconstruction error:

$$\underset{\mathbf{W} \in \mathbb{R}^{d \times \ell}}{\operatorname{argmin}} \sum_{i=1}^{n} \left\| \mathbf{a}_i - \mathbf{W}\mathbf{W}^\top \mathbf{a}_i \right\|^2 \tag{34}$$

where $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_\ell$. The columns of $\mathbf{W}$ are ℓ orthonormal directions spanning the chosen subspace. Multiplying by $\mathbf{W}^\top$ gives each data point's coordinates along these directions, and multiplying again by $\mathbf{W}$ maps those coordinates back to the original space. Thus $\mathbf{W}\mathbf{W}^\top \mathbf{A}$ is the reconstruction of our data we get using only the information captured by our subspace.

Since $\mathbf{W}$ has rank ℓ, the reconstruction retains only ℓ dimensions. For $\ell = 1$ (i.e., to find the first principal component), this can be rewritten as

$$\underset{\mathbf{u} \in \mathbb{R}^d}{\operatorname{argmin}} \sum_{i=1}^{n} \left\| \mathbf{a}_i - \langle \mathbf{a}_i, \mathbf{u} \rangle \mathbf{u} \right\|^2 \tag{35}$$

where $\|\mathbf{u}\| = 1$. **Show that finding the top principal component is equivalent to maximizing**

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \tag{36}$$

Answer: We can begin by writing out the norm as an inner product:

$$\sum_{i=1}^{n} \left\| \mathbf{a}_i - \langle \mathbf{a}_i, \mathbf{u} \rangle \mathbf{u} \right\|^2 \tag{37}$$

$$= \sum_{i=1}^{n} \left(\mathbf{a}_i - \langle \mathbf{a}_i, \mathbf{u} \rangle \mathbf{u} \right)^\top \left(\mathbf{a}_i - \langle \mathbf{a}_i, \mathbf{u} \rangle \mathbf{u} \right) \tag{38}$$

$$= \sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{a}_i \rangle - 2 \langle \mathbf{a}_i, \mathbf{u} \rangle^2 + \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \underbrace{\langle \mathbf{u}, \mathbf{u} \rangle}_{1} \tag{39}$$

$$= \sum_{i=1}^{n} \|\mathbf{a}_i\|^2 - \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \tag{40}$$

We cannot change $\|\mathbf{a}_i\|^2$ (the data points are fixed), so minimizing this expression is the same as minimizing

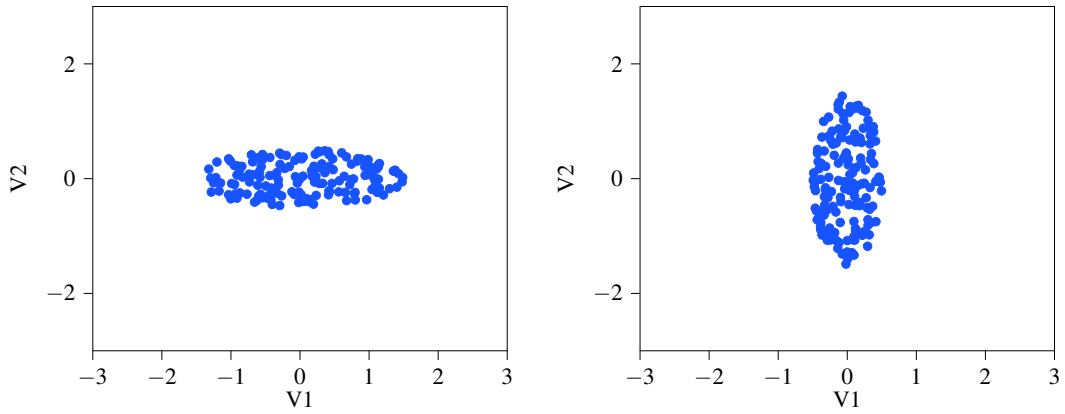
$$\sum_{i=1}^{n} \left(- \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \right) = - \sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \tag{41}$$

which is the same as maximizing

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \tag{42}$$

In each plot below, the data is projected onto two unit vectors. The x coordinate is the projection onto the first vector (written as "V1" or $\mathbf{v}_1$), and the y coordinate is the projection onto the second vector (written as "V2" or $\mathbf{v}_2$). We say that a plot is "valid" if the first vector would be the first principal component, and if the second vector would correspond to the second principal component. For each subpart, explain your answer.

(b) **Which of these two plots is valid?**



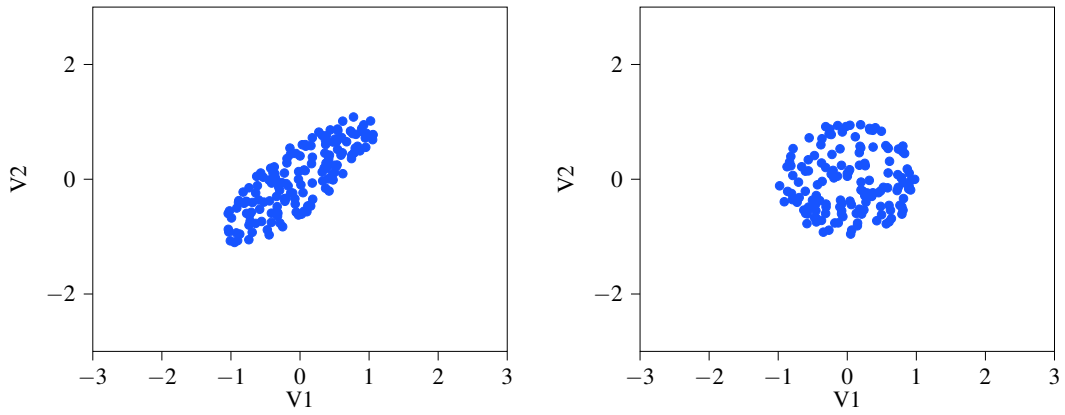
Answer: To check if a plot is valid, we can see whether the $\mathbf{v}_1$ satisfies the definition of the first principal component, namely that it should maximize

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{v}_1 \rangle^2 \tag{43}$$

In other words, this is the same as maximizing the sum of squares of the x-coordinates in the plots above.

In this case, the plot on the left has the largest range of values, from -2 to 2. This is larger than the spread along the y-axis, so the sum of squares of x-coordinates is maximized. Hence, the left plot is valid. For the right plot, there exists another direction (i.e., the direction of $\mathbf{v}_2$) that maximizes the sum of squares, so $\mathbf{v}_1$ cannot be the first principal component.

(c) **Which of these two plots is valid?**



Answer: The plot on the left has maximum spread along the direction spanned by $\frac{1}{\sqrt{2}}\mathbf{v}_1 + \frac{1}{\sqrt{2}}\mathbf{v}_2$, which is not the same as $\mathbf{v}_1$. In other words,

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{v}_1 \rangle^2 \neq \operatorname*{argmax}_{\mathbf{u} \in \mathbb{R}^d} \sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2 \tag{44}$$

Hence, $\mathbf{v}_1$ is not the first principal component and the plot is invalid. For the second plot, the sum of squares of x-coordinates appears to be the same as the sum of squares along any other axis, so it is entirely plausible for $\mathbf{v}_1$ to be the first principal component.

(d) Part (a) showed that finding the first principal component means maximizing $\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u} \rangle^2$ over unit vectors $\mathbf{u}$. We will now connect this to the SVD of the data matrix.

Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the SVD of the data matrix, with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and left singular vectors $\mathbf{u}_1, \ldots, \mathbf{u}_r$. **Show that evaluating the objective above at $\mathbf{u} = \mathbf{u}_k$ gives**

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u}_k \rangle^2 = \sigma_k^2. \tag{45}$$

Then explain what this says about which vector is the first principal component.

Hint: The sum is over the columns $\mathbf{a}_i$ of $\mathbf{A}$, so it collects the entries of the row vector $\mathbf{u}_k^\top \mathbf{A}$. What is $\mathbf{u}_k^\top \mathbf{A}$ in terms of the SVD?

Answer: Collecting the terms of the sum into a single row vector,

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u}_k \rangle^2 = \sum_{i=1}^{n} \left(\mathbf{u}_k^\top \mathbf{a}_i \right)^2 = \left\| \mathbf{u}_k^\top \mathbf{A} \right\|^2, \tag{46}$$

since the ith entry of $\mathbf{u}_k^\top \mathbf{A}$ is exactly $\mathbf{u}_k^\top \mathbf{a}_i$. Substituting the SVD and using the fact that the columns of $\mathbf{U}$ are orthonormal, so that $\mathbf{u}_k^\top \mathbf{U} = \mathbf{e}_k^\top$,

$$\mathbf{u}_k^\top \mathbf{A} = \mathbf{u}_k^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{e}_k^\top \mathbf{\Sigma}\mathbf{V}^\top = \sigma_k \mathbf{v}_k^\top. \tag{47}$$

Therefore

$$\sum_{i=1}^{n} \langle \mathbf{a}_i, \mathbf{u}_k \rangle^2 = \left\| \sigma_k \mathbf{v}_k^\top \right\|^2 = \sigma_k^2 \left\| \mathbf{v}_k \right\|^2 = \sigma_k^2, \tag{48}$$

where the last step uses $\|\mathbf{v}_k\| = 1$.

What this means. The quantity we are trying to maximize, evaluated at the kth left singular vector, is exactly σ_k^2. So among the $\mathbf{u}_k$, the best direction to project onto is the one paired with the largest singular value. Since we write the singular values in descending order, that is $\mathbf{u}_1$: the first principal component is the first column of $\mathbf{U}$.

It is worth being careful about what is doing the work here. The data determines which direction captures the most, and the objective at $\mathbf{u}_k$ is σ_k^2 no matter how we choose to write the SVD down. Listing the singular values in descending order is a convention, and it is what makes the best direction come out with the index 1 rather than some other index. The convention is useful precisely because it makes "the first ℓ columns of $\mathbf{U}$" and "the best ℓ-dimensional subspace to project onto" the same thing, which is also why truncating an SVD after the first few terms gives a good low-rank approximation.

One caveat: if $\sigma_1 = \sigma_2$, then (45) gives the same value for $\mathbf{u}_1$ and $\mathbf{u}_2$, and in fact every unit vector in their span does equally well. In that case there is no unique first principal component.

6. Movie Ratings and PCA

For this problem, imagine that we can think of movie ratings as a structured set of data. For every person i and movie j, we have that person's rating $\mathbf{R}_{ij}$ (thought of as a real number).

Suppose that there are m movies and n people. Let's think about arranging this data into a big $n \times m$ matrix $\mathbf{R}$ with people corresponding to rows and movies corresponding to columns. So the entry in the i-th row and j-th column should be $\mathbf{R}_{ij}$ above. Note that this is organized differently from how it was in lecture. Each row corresponds to a unique person and each column to a unique movie.

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} & \cdots & \mathbf{R}_{1m} \\ \mathbf{R}_{21} & \mathbf{R}_{22} & \cdots & \mathbf{R}_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{R}_{n1} & \mathbf{R}_{n2} & \cdots & \mathbf{R}_{nm} \end{bmatrix} \tag{49}$$

(a) Suppose we believe that there is actually an underlying pattern to this rating data and that a separate study has revealed that every movie is characterized by a set of characteristics: say action and comedy. This means that every movie j has a pair of numbers a_j (for action) and c_j (for comedy) that define it. At the same time, every person i has a pair of sensitivities f_i and g_i that define their preferences for action vs. comedy in movies respectively. A person i will rate the movie j as $\mathbf{R}_{ij} = f_i a_j + g_i c_j$.

If we arrange the sensitivities into a pair of n-dimensional vectors $\mathbf{f}, \mathbf{g}$ for our group of n people, and the movie characteristics into a pair of m-dimensional vectors $\mathbf{a}, \mathbf{c}$ for our group of m movies, **use outer products to express the rating matrix R in terms of these vectors $\mathbf{f}, \mathbf{g}, \mathbf{a}, \mathbf{c}$.**

Solution:

$$\begin{aligned} \mathbf{R} &= \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} & \cdots & \mathbf{R}_{1m} \\ \mathbf{R}_{21} & \mathbf{R}_{22} & \cdots & \mathbf{R}_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{R}_{n1} & \mathbf{R}_{n2} & \cdots & \mathbf{R}_{nm} \end{bmatrix} \\ &= \begin{bmatrix} f_1 a_1 + g_1 c_1 & f_1 a_2 + g_1 c_2 & \cdots & f_1 a_m + g_1 c_m \\ f_2 a_1 + g_2 c_1 & f_2 a_2 + g_2 c_2 & \cdots & f_2 a_m + g_2 c_m \\ \vdots & \vdots & \cdots & \vdots \\ f_n a_1 + g_n c_1 & f_n a_2 + g_n c_2 & \cdots & f_n a_m + g_n c_m \end{bmatrix} \\ &= \begin{bmatrix} f_1 a_1 & f_1 a_2 & \cdots & f_1 a_m \\ f_2 a_1 & f_2 a_2 & \cdots & f_2 a_m \\ \vdots & \vdots & \cdots & \vdots \\ f_n a_1 & f_n a_2 & \cdots & f_n a_m \end{bmatrix} + \begin{bmatrix} g_1 c_1 & g_1 c_2 & \cdots & g_1 c_m \\ g_2 c_1 & g_2 c_2 & \cdots & g_2 c_m \\ \vdots & \vdots & \cdots & \vdots \\ g_n c_1 & g_n c_2 & \cdots & g_n c_m \end{bmatrix} \\ &= \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} + \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{bmatrix} \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix} \\ &= \mathbf{f}\mathbf{a}^\top + \mathbf{g}\mathbf{c}^\top \end{aligned} \tag{50, 51}$$

(b) Now suppose that we want to discover the underlying nature of movies from the data $\mathbf{R}$ itself.

Suppose for this part that you have four observed rating data vectors (corresponding to four different movies being rated by six individuals).

All of the movie data vectors just happened to be multiples of the following 6-dimensional vector

$$\mathbf{w} = \begin{bmatrix} 2 \\ -2 \\ 3 \\ -4 \\ 4 \\ 0 \end{bmatrix}.$$

(For your convenience, note that $\|\mathbf{w}\| = 7$.)

You arrange the movie data vectors as the columns of a matrix $\mathbf{R}$ given by:

$$\mathbf{R} = \begin{bmatrix} | & | & | & | \\ -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} \\ | & | & | & | \end{bmatrix} \tag{52}$$

You want to perform PCA (for movies) using the SVD of the matrix $\mathbf{R}$ to better understand the pattern in your data.

The first "principal component vector" is a *unit vector* that tells which direction we would want to project the columns of $\mathbf{R}$ onto to get the best rank-1 approximation for $\mathbf{R}$.

Find this first principal component vector of the columns of R to explain the nature of your movie data points.

(HINT: You don't need to actually compute any SVDs in this simple case. Also, be sure to think about what size vector you want as the answer. Don't forget that you want a unit vector!)

Solution: Principal component analysis is in general about understanding how best to approximate our (potentially) high-dimensional data (like recordings from a microphone, or in this case, a movie's ratings by lots of people) with its lower-dimensional essence. The first principal component is about seeing which one-dimensional line best approximates the data points — i.e. which is the line for which projecting the data points onto it results in "estimates" that are as close as possible to the data points.

In the case of this problem, every point is explicitly given as a multiple of a single vector $\mathbf{w}$ and so the data already lies on such a straight line going through the origin. So, the first principal component is just along the direction of $\mathbf{w}$. Because a principal component represents a direction, it is conventional to normalize the vector to have unit length. In this case, we are told that the vector $\mathbf{w}$ has length 7, and so the answer is $\frac{\mathbf{w}}{7}$. (Because the line is all that matters, you could also have used the negative of this $-\frac{\mathbf{w}}{7}$.)

We can also do this using the SVD.

The singular value decomposition of a matrix $\mathbf{R}$ is a way of decomposing $\mathbf{R}$ into a sum of rank 1 matrices. In this sum the ith rank 1 matrix is formed from taking the outer product of normalized column vectors $\mathbf{u}_i$ and normalized row vectors $\mathbf{w}_i^\top$, scaled by their respective singular values σ_i.

Looking at our given $\mathbf{R}$, we can see that the matrix itself is rank 1 as the columns are all multiples of the same vector: $\mathbf{w}$. Seeing this we realize we can rewrite the matrix $\mathbf{R}$ as the following outer product:

$$\mathbf{R} = \begin{bmatrix} | \\ \mathbf{w} \\ | \end{bmatrix} \begin{bmatrix} -1 & -2 & 2 & 4 \end{bmatrix} \tag{53}$$

However the SVD requires we normalize the vectors $\mathbf{u_1}$ and $\mathbf{v_1^\top}$. In order to reconstruct A properly we must scale back with the norms that we divided out to normalize.

$\|[-1,-2,2,4]\| = \sqrt{25} = 5$ and $\|\mathbf{w}\| = 7$. Consequently, when we pull that out, we get $\sigma_1 = 35$ as the singular value that corresponds to the first (and only) principal component.
Thus we can write the SVD of $\mathbf{R}$ as:

$$\mathbf{R} = \begin{bmatrix} | \\ \frac{\mathbf{w}}{7} \\ | \end{bmatrix} 35 \begin{bmatrix} \frac{-1}{5} & \frac{-2}{5} & \frac{2}{5} & \frac{4}{5} \end{bmatrix} \tag{54}$$

Now we just have to pick which normalized vector to deem the principal component. Since our data (the movie ratings) are collected as columns we choose $\frac{\mathbf{w}}{7}$ as the principal component.

Alternate Solution:
To find the first principal component along the columns, we can use $\mathbf{\Sigma}$ and $\mathbf{U}$. This is because our data is stored in the columns of $\mathbf{R}$. We know that

$$\mathbf{R} = \mathbf{U\Sigma V^\top} \tag{55}$$

$$\implies \mathbf{RR^\top} = \mathbf{U\Sigma\Sigma^\top U^\top} \tag{56}$$

where $\mathbf{\Sigma\Sigma^\top}$ represents the square diagonal matrix with $min(m,n)$ singular values squared on the diagonal. Plugging in for $\mathbf{R}$ gives

$$\mathbf{RR^\top} = \begin{bmatrix} -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} \end{bmatrix} \begin{bmatrix} -\mathbf{w} \\ -2\mathbf{w} \\ 2\mathbf{w} \\ 4\mathbf{w} \end{bmatrix} \tag{57}$$

$$\mathbf{RR^\top} = \mathbf{ww^\top} + 4\mathbf{ww^\top} + 4\mathbf{ww^\top} + 16\mathbf{ww^\top}$$

$$\mathbf{RR^\top} = 25\mathbf{ww^\top} \tag{58}$$

By the Spectral Theorem, we know that the eigenvectors of $\mathbf{RR^\top}$ correspond to the columns of $\mathbf{U}$ and the eigenvalues of $\mathbf{RR^\top}$ correspond to the singular values squared. By inspection, we can see that the eigenvector is $\mathbf{w}$. Solving for the eigenvalue and singular value:

$$\mathbf{RR^\top w} = 25\mathbf{ww^\top w} \tag{59}$$

$$\mathbf{RR^\top w} = (25\mathbf{w^\top w})\mathbf{w}$$

$$\implies \lambda = 25\mathbf{w^\top w} = 5^2 7^2$$

$$\implies \sigma = \sqrt{\lambda} = 35 \tag{60}$$

For PCA, we require normalized vectors, so for that reason our first principal component is $\frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{\mathbf{w}}{7}$ with a corresponding singular value of $\sigma = 35$.

(c) Suppose that we now collected two more data points (corresponding to two more movies being rated by the same set of six people, i.e. two additional columns for our matrix) that are multiples of a different vector $\mathbf{p}$ where:

$$\mathbf{p} = \begin{bmatrix} 6 \\ 3 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

(For your convenience, note that $\|\mathbf{p}\| = 7$ and that $\mathbf{p}^\top \mathbf{w} = 0$.)

We augment our ratings data matrix with these two new data points to get:

$$\mathbf{R} = \begin{bmatrix} | & | & | & | & | & | \\ -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} & -3\mathbf{p} & 3\mathbf{p} \\ | & | & | & | & | & | \end{bmatrix} \tag{61}$$

Find the first two principal components corresponding to the nonzero singular values of $\mathbf{R}$. This is what we would use to best project the movie data points onto a two-dimensional subspace.

What is the first principal component vector? What is the second principal component vector? Justify your answer. *(HINT: Think about the inner product of* $\mathbf{w}$ *and* $\mathbf{p}$ *and what that implies for being able to appropriately decompose* $\mathbf{R}$*. Again, very little computation is required here.)*

Solution: The solution to the previous part tells you what we need to do. We need to find the best two-dimensional subspace that best represents our data.

We start by taking the SVD of $\mathbf{R}$.

The columns of $\mathbf{R}$ are all multiples of two vectors: $\mathbf{w}$ and $\mathbf{p}$. Each of these can be used to create a rank 1 matrix, and these can be summed together to form $\mathbf{R}$.

Since $\mathbf{w}$ and $\mathbf{p}$ are orthogonal to one another, our life is easier. This problem's $\mathbf{R}$ matrix is made especially nice by seeing that a data point is either purely in the $\mathbf{w}$ direction, or the $\mathbf{p}$ direction.

Using this knowledge we rewrite $\mathbf{R}$ as:

$$\mathbf{R} = \begin{bmatrix} | \\ \mathbf{w} \\ | \end{bmatrix} \begin{bmatrix} -1 & -2 & 2 & 4 & 0 & 0 \end{bmatrix} + \begin{bmatrix} | \\ \mathbf{p} \\ | \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & -3 & 3 \end{bmatrix}. \tag{62}$$

The orthogonality relationships demanded by the SVD are clearly satisfied since the row-vectors involved above have disjoint support (i.e. when one is nonzero, the other is zero) and the columns are orthogonal since we've been told so.

However for the SVD the vectors: $\mathbf{u_1}$, $\mathbf{v_1}^\top$, $\mathbf{u_2}$ and $\mathbf{v_2}^\top$ must be normalized and each rank 1 matrix must be scaled by the appropriate σ_i to allow the sum to properly reconstruct $\mathbf{R}$. We also need to figure out which σ_i is bigger so we can order them properly. In the previous part, we have already done the calculations for $\mathbf{w}$'s part in this story. So what remains is the $\mathbf{p}$ part. Clearly the norm of the relevant row is $3\sqrt{2}$ which the norm of the relevant column is 7. So the singular value in question is $21\sqrt{2}$.

Using this we can rewrite $\mathbf{R}$ as:

$$= \begin{bmatrix} | \\ \frac{\mathbf{w}}{7} \\ | \end{bmatrix} 35 \begin{bmatrix} \frac{-1}{5} & \frac{-2}{5} & \frac{2}{5} & \frac{4}{5} & 0 & 0 \end{bmatrix} + \begin{bmatrix} | \\ \frac{\mathbf{p}}{7} \\ | \end{bmatrix} 21\sqrt{2} \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{-3}{3\sqrt{2}} & \frac{3}{3\sqrt{2}} \end{bmatrix}. \tag{63}$$

Since $35 > 21\sqrt{2}$, we arrange them in descending order such that our singular values are $\sigma_1 = 35$ and $\sigma_2 = 21\sqrt{2}$. Thus $\frac{\mathbf{w}}{7}$ which corresponds to σ_1 is still the first principal component vector and $\frac{\mathbf{p}}{7}$ which

corresponds to σ_2 is the second principal component vector.

Alternate Solution:

Again, to find the first principal component along the columns, we can use $\mathbf{\Sigma}$ and $\mathbf{U}$. This is because our data is stored in the columns of $\mathbf{R}$. We know that

$$\mathbf{R} = \mathbf{U\Sigma V}^\top \tag{64}$$

$$\implies \mathbf{RR}^\top = \mathbf{U\Sigma\Sigma}^\top\mathbf{U}^\top \tag{65}$$

where $\mathbf{\Sigma\Sigma}^\top$ represents the square diagonal matrix with $min(m,n)$ singular values squared on the diagonal. Plugging in for $\mathbf{R}$ gives

$$\mathbf{RR}^\top = \begin{bmatrix} -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} & -3\mathbf{p} & 3\mathbf{p} \end{bmatrix} \begin{bmatrix} -\mathbf{w} \\ -2\mathbf{w} \\ 2\mathbf{w} \\ 4\mathbf{w} \\ -3\mathbf{p} \\ 3\mathbf{p} \end{bmatrix} \tag{66}$$

$$\mathbf{RR}^\top = \mathbf{ww}^\top + 4\mathbf{ww}^\top + 4\mathbf{ww}^\top + 16\mathbf{ww}^\top + 9\mathbf{pp}^\top + 9\mathbf{pp}^\top \tag{67}$$

$$\mathbf{RR}^\top = 25\mathbf{ww}^\top + 18\mathbf{pp}^\top \tag{68}$$

By the spectral decomposition, we know that the eigenvectors of $\mathbf{RR}^\top$ correspond to the columns of $\mathbf{U}$ and the eigenvalues of $\mathbf{RR}^\top$ correspond to the singular values squared. By inspection, we can see that the first eigenvector is $\mathbf{w}$ and the second one is $\mathbf{p}$. Starting with the first:

$$\mathbf{RR}^\top\mathbf{w} = 25\mathbf{ww}^\top\mathbf{w} + 18\mathbf{p}(\mathbf{p}^\top\mathbf{w}) \tag{69}$$

$$\mathbf{RR}^\top\mathbf{w} = (25\mathbf{w}^\top\mathbf{w})\mathbf{w} + 0 \tag{70}$$

$$\implies \lambda_1 = 25\mathbf{w}^\top\mathbf{w} = 5^2 7^2 \tag{71}$$

$$\implies \sigma_1 = \sqrt{\lambda_1} = 35 \tag{72}$$

When computing the first eigenvalue, we used the fact that $\mathbf{p}^\top\mathbf{w} = 0$. We can repeat the same process for the second eigenvalue:

$$\mathbf{RR}^\top\mathbf{p} = 25\mathbf{w}(\mathbf{w}^\top\mathbf{p}) + 18\mathbf{p}(\mathbf{p}^\top\mathbf{w}) \tag{73}$$

$$\mathbf{RR}^\top\mathbf{p} = 0 + 18(\mathbf{p}^\top\mathbf{p})\mathbf{p} \tag{74}$$

$$\implies \lambda_2 = 18\mathbf{p}^\top\mathbf{p} = 3^2 7^2 2 \tag{75}$$

$$\implies \sigma_2 = \sqrt{\lambda_2} = 21\sqrt{2} \tag{76}$$

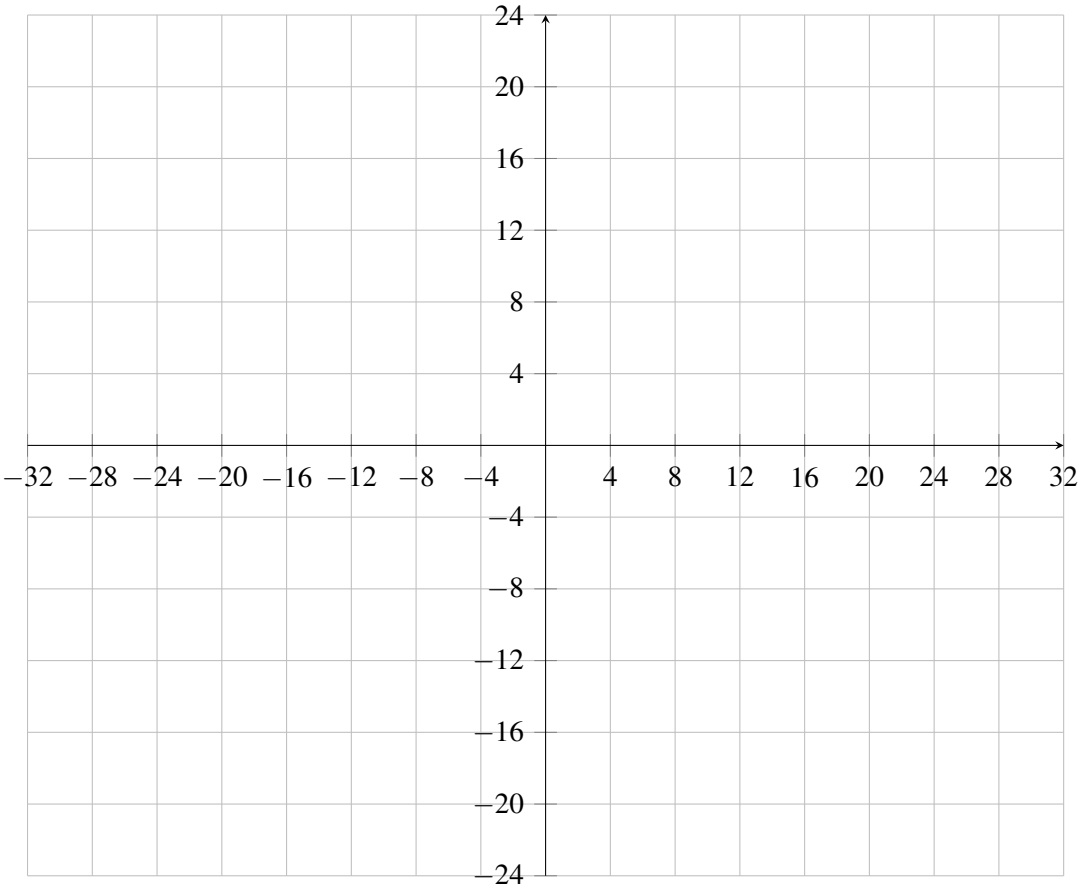
For PCA, we require normalized vectors, so for that reason our first principal component is $\frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{\mathbf{w}}{7}$ with a corresponding singular value of $\sigma_1 = 35$. Our second principal component is $\frac{\mathbf{p}}{\|\mathbf{p}\|} = \frac{\mathbf{p}}{7}$ with a corresponding singular value of $\sigma_2 = 21\sqrt{2}$. As mentioned in the solution above, we choose $\sigma_1 = 35$, $\sigma_2 = 21\sqrt{2}$ in this order because $35 > 21\sqrt{2}$.

(d) In the previous part, you had

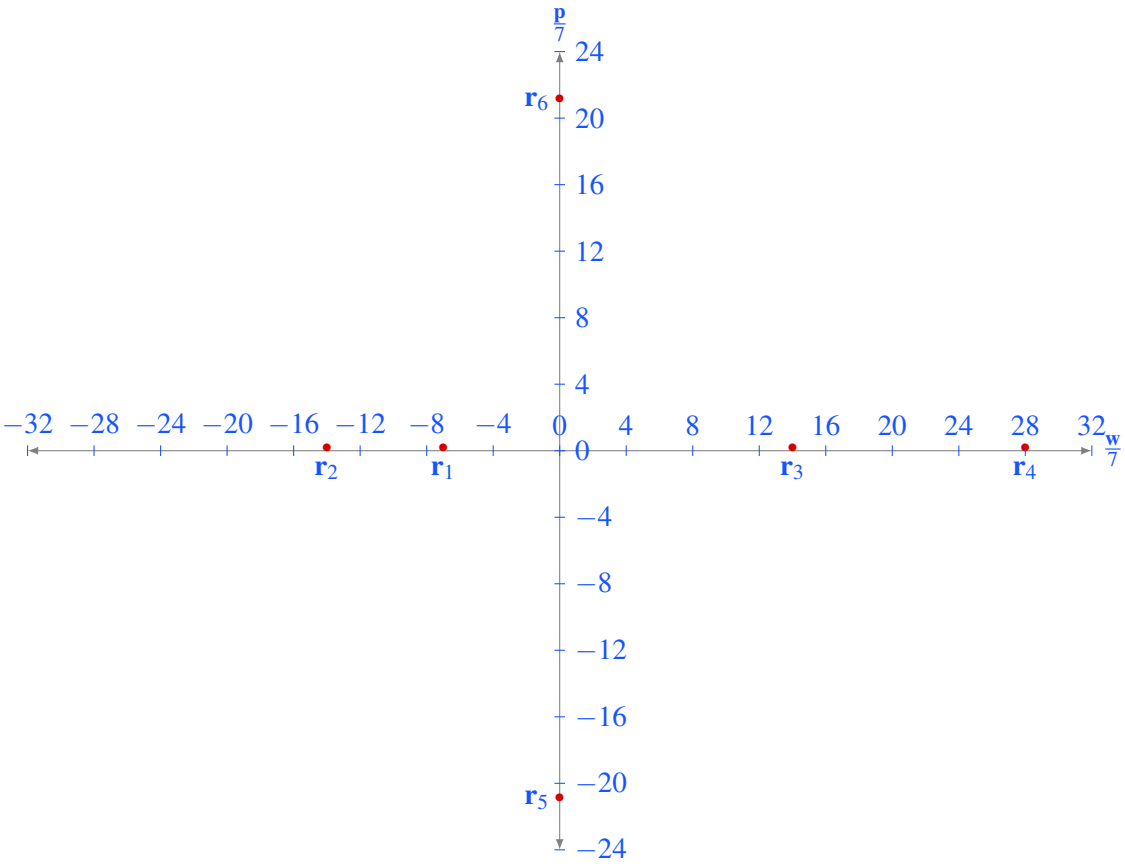
$$\mathbf{R} = \begin{bmatrix} | & | & | & | & | & | \\ -\mathbf{w} & -2\mathbf{w} & 2\mathbf{w} & 4\mathbf{w} & -3\mathbf{p} & 3\mathbf{p} \\ | & | & | & | & | & | \end{bmatrix}$$

with $\|\mathbf{w}\| = 7$ and $\|\mathbf{p}\| = 7$, satisfying $\mathbf{p}^\top \mathbf{w} = 0$.

Plot the movie data points $\mathbf{r}_i$ (for all i, where $\mathbf{r}_i$ denotes the i-th column of R) projected onto the first and second principal component vectors of the columns of R. The horizontal axis should reflect the coordinate along the first principal component, and the vertical axis should reflect the coordinate along the second principal component. **Label each point, and the axes. Remember that principal component vectors are normalized**.



Solution: Once we know what the principal components are, we know that the first four data points are just multiples of the first principal component and the last two data points are just multiples of the second principal component. What multiples? For the first four, the multiples are clearly $-7, -14, 14, 28$ since the norm of $\mathbf{w}$ is 7. For the final two, the multiples are clearly $-21, +21$ since the norm of $\mathbf{p}$ is also 7. Plotting:


p/7
24
20
16
12
8
4
r6
−32 −28 −24 −20 −16 −12 −8 −4 0 4 8 12 16 20 24 28 32
w/7
r2
r1
r3
r4
−4
−8
−12
−16
−20
r5
−24

7. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.